सहायक प्राध्यापक (जीवभौतिकशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
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परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११६९)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Molecular Biophysics:** Biomolecules and biopolymers- amino acids, their classification and properties, peptides, nucleotides and their properties, oligonucleotides, monosaccharides, polysaccharides and their properties, linkages in polysaccharides, structure and function of cellulose and chitin, glyco-conjugates, classification and biological role of vitamins and hormones. Nucleic acids- structure of DNA, conformational parameters of nucleic acids and their constituents, Chargaff's rule, DNA polymorphism, DNA supercoiling, hyperchromicity, circular DNA, types and structure of mRNA, tRNA, rRNA and modified nucleotides, nucleic acid sequencing. Proteins- structural levels of proteins and stabilizing factors, protein conformation, Ramchandran plot, helical parameters and conformation, organization and interaction of angles, structure of alpha keratin, silk fibrin, collagen, actin, myosin, haemoglobin, myoglobin, lysozyme, cytochromes, classification and role of beta-bends and bulges, super secondary structure, domains and motifs, proteins in solution, protein sequencing, concept of protein evolution with reference to cytochrome and haemoglobin.
2	**Biophysical chemistry:** Water, acids, bases and redox potential- structure of water molecule, association of water molecules through hydrogen bonding, nature of hydrophobic interactions, physic-chemical properties of water, state of water in biological systems and its significance, strong and weak acids and bases, molarity and normality, amphoteric electrolytes, pH, determination of pH from concentration of H^+ and OH^- ions, measurement of pH, Henderson and Haselbach equation, pH titration curve, pKa value and its significance, buffers, oxidation and reduction, redox potential, equivalence and electrical and chemical energy, electrochemical cells, galvanic cell, potential of half-cell, Nernst equation, examples of redox potential in biological systems.

	Thermodynamics and bioenergetics- laws of thermodynamics, concept of free-energy, unavailable energy and entropy, negative entropy change in living systems, heat content of food, bomb calorimeter, energy generation and energytransfer processes in biochemical reactions, metabolism of glucose and generation of ATP, energy requirements in cell metabolism, role and structure of mitochondria, high-energy phosphate bond, electron transfer phenomenon and biological energy transfer. Macromolecular interactions- ligand interaction at equilibrium, identical independent sites, Scatchard plot, multiple classes of independent sites, interaction between binding sites, allosterism, MWC model, sequential model, binding of oxygen to haemoglobin, binding of two different ligands, energetics and dynamics of binding, structures of protein-ligand complexes, relationship between protein conformation and binding, binding of immunoglobulins and DNA-binding proteins, free radicals in biology and medicine.
3	**Cellular Biophysics:** Membrane structure and function- models of membrane, composition, electrical properties, principles of membrane organization and stability, biogenesis of cell membrane, self-organization of lipids and proteins- linear aggregates of membrane components, aggregation of lipid molecules in mycelia and lipid double layers, biologically important non-lamellar lipid phases, formation of flexible membranous nanodomains (lipid rafts), lateral phase separation of membranous components, aggregation of nanodomains, formation and stability of membranous nanotubes. Elastic properties of membranes- deformation in levels of a membrane, flexible energy, influence of forms of membrane components and direct interactions between membranous components in elastic properties of membranes, lateral distribution of membranous components and elastic properties of membranes, elastic properties of membranes and forms of cells and organelles, influence of cytoskeleton on forms of cells, molecular motion in membrane and membrane fluidity, protein-lipid interactions, phase properties of biological membranes. Cell-cell interactions- connection between cell and its environment, glycocalyx, extracellular matrix, collagen, elastin, fibronectin, lamin, proteoglycans, integrins, cell junctions, desmosomes, gap junction, tight junction, plasmodesmata, synapse and synaptic vesicles, cell signalling, paracrine, autocrine, endocrine and synaptic signalling, heat shock proteins, G-protein structure and role in signalling, cyclic AMP, role of calcium ions in signalling, CAM kinases, interaction between cyclic MP and Ca^{++} ions, role of methylation in adaptation and bacterial chemotaxis.
4	**Microscopic Techniques:** General introduction- electromagnetic radiations, properties of light, light intensity, amplitude, frequency, wavelength, polarization, microscope design, parts of microscope, magnification, resolution, numerical aperture, principles of light microscopy, types of light microscope, image formation, aberrations, imaging modes, specimen preparation, bright field microscopy, dark field microscopy, phase-contrast microscopy, applications and limitations of light microscopy. Electron microscopy- introduction to transmission electron microscopy, instrumentation, electron sources, electromagnetic lenses, imaging modes, image contrast, bright field, dark field, magnification, resolution, energy filtered TEM, electron diffraction, techniques in TEM, negative staining, shadow casting, freeze fracture, freeze etching, applications and limitations of TEM, introduction to scanning electron microscopy, instrumentation, electron beam-specimen interaction, energy dispersive spectroscopy (EDS), types of signals produced by SEM- x-rays, light rays, secondary electrons, back scattered electrons, applications and limitations of SEM. Scanning probe microscopy- general introduction, scanning tunnelling microscopy, atomic force microscopy, principle of working, working modes, applications and limitations of scanning tunnelling microscopy and atomic force microscopy. Advanced microscopy techniques- fluorescence microscopy (principle, design and working), choosing a probe, choosing a light source, choosing a filter set, controls for antibody staining, bleed through, autofluorescence, FRET, fluorophores, types of fluorescence microscopes, epifluorescence microscopy, confocal microscope, multiphoton microscope, total internal reflection fluorescence (TIRF), applications and limitations of fluorescence microscopy.
5	**Molecular enzymology:** Enzymes as biocatalysts- remarkable properties of enzymes as biocatalysts, active site of enzymes, three-point attachment, mechanism of enzyme action, flexible enzymes, lock-and-key

	hypothesis, induced-fit hypothesis, catalytic efficiency of enzymes, micro-environmental approach to enzyme dynamics, enzyme nomenclature and classification, different classes of enzymes with examples, isomerization, rearrangement and condensation reactions, molecular dynamics and transient state in enzyme catalysis. Basic principles of chemical catalysis and enzyme kinetics- introduction to chemical kinetics, velocity, order and molecularity of a chemical reaction, kinetic equations for zero, first, second and third order reactions, determination of order of reaction, Arrhenius equation, activation energy and its estimation, collision and transition state theories of reaction rate, catalysts, mode of action of catalysts, nucleophilic, electrophilic and acid-base catalysis, enzyme catalysis, kinetics of single substrate reaction, Michaelis equation, steady state kinetics, transient phases of enzyme reactions, Lineweaver-Burk plot, Eddie-Hofstee plot, Woolf plot, effect of pH, temperature, metal ions and substrate concentration on enzyme activity, enzyme turn over, mechanisms of multi-substrate enzyme reactions (conceptual approach), kinetics of reversible enzyme inhibition, mechanism of action of chymotrypsin and ribonuclease. Enzyme regulation and enzyme technology- control of enzyme activity, feedback inhibition, kinetic behaviour of allosteric enzymes, mechanism of allosteric interactions, subunit structure and protein assembly, aspartate transcarbamoylase, proton ATPase, metalloenzymes, carboxypeptidase A, role of zinc in enzyme activity, enzyme immobilization techniques, use of isolated enzymes in industrial processes, enzymes in clinical diagnosis, isozymes, abzymes, ribozymes, enzyme therapy, extremozymes, solventogenic and non-aqueous enzymes.
6	**Environmental Biophysics:** Introduction to environment, microenvironment, energy exchange, mass and momentum transport, conservation of energy and mass, continuity in the biosphere. Atmospheric temperature and its behaviour, temperature variation, modelling of vertical and temporal variation in air temperature, thermal time, temperature and biological development. Water potential in organism and their surrounding. General understanding of conductance of heat, mass and transport; molecular diffusion, diffusive conductance of integument, conductance of heat, mass and transport in laminar force convection. Animal shapes and heat conductance. Radiation fluxes in natural environment, direct, diffuse and reflected radiation, direct and diffuse shortwave irradiance, solar radiation under clouds, absorptivities for thermal and solar radiation. Energy concept, metabolism, conduction of heat in animal coats and tissues, quantitative analysis of animal thermal response, operative temperature, complexities of animal energetic, animal and water; human and their environment: metabolism, evaporation, survival in cold and heat. Leaf temperature, aerodynamics temperature of plant canopies, transpiration, canopy transpiration, assimilation models, biochemical model, stomatal conduction, leaf area index and light transmission, light transmission through stomatal conductance, transmission of diffuse radiation, light scattering in canopies, reflection of light in plants, light scattering in canopies, reflection of light by plant.
7	**Physiology and Biophysics:** Brain, neurophysiology and special senses- general anatomy of brain, central and peripheral nervous system, nerve cell structure, myelinated and non-myelinated nerve cells, blood brain barrier, generation of nerve impulse, synaptic transmission, physic-chemical basis of membrane potential, resting and action potential, propagation of action potential, voltage clamp, patch clamp technique, Hadgkin-Huxley analysis, motor and cortical control, sleep and consciousness, neuromuscular junction, excitation contraction coupling, neuronal networks, processing of information, memory and neuropeptides, special senses- biophysics of sensory mechanism and function of receptor cells, cutaneous, olfactory and gustatory sensations, vision, physical aspects of vision- neurophysiology of color vision, visual evoked potentials, audition- physical aspects, auditory transduction, acoustic encoding. Cardiovascular and pulmonary physiology- physical characteristics of blood, hemodynamics principles and equations, genesis and spread of cardiac impulse, cardiodynamics, regulation of blood pressure and blood volume, heart rate, cardiac output and venous return, cardiovascular responses to stress (exercise, shock and hypertension), biophysical aspects of lung expansion, respiratory mechanics and gas exchange process, gas diffusion and transport, pulmonary function test and its significance.

	Renal and reproduction physiology- ionic composition and distribution of body fluids, body fluid osmolality, dialysis and dehydration, biophysical aspects of renal filtration and blood flow, renal tubules and their function, effective circulation volume, autoregulation, reabsorption and secretion, regulation of acid-base balance, hormonal control of reproductive mechanisms, morphology and dynamics of sperms, kinematics parameters of sperm movement and sperm motility, basic principles of assisted reproductive technology- IUI and IVF techniques.
8	**Molecular Biology and Genetics:** Genome organization- Mendelian genetics, extra-chromosomal inheritance, polygenic inheritance, heritability and its measurements, QTL mapping, nucleic acids as genetic material, topology of nucleic acid and role of topoisomerases, concept of gene, chromosome organization in prokaryotes, eukaryotes and viruses, spatial arrangement and role of histones, chromatin, nucleosomes, solenoid model, loops, domains and scaffolds in chromatin, Law of DNA constancy, C-value paradox, repetitive DNA, selfish DNA, split genes, pseudogenes, cryptic genes, promiscuous DNA, multigene families, DNA replication in prokaryotes and eukaryotes- mechanisms and enzymes involved. Gene expression and regulation- central dogma of molecular biology, genetic code, salient features of genetic code, protein synthesis- mechanism of transcription in prokaryotes and eukaryotes, m-RNA synthesis and processing (capping, splicing, RNA editing, polyadenylation), role of ribozymes, ribosome structure and function, Shine-Dalgarno sequence, TATA box, termination of transcription, mechanism of translation, formation of aminoacyl t-RNA, initiation, elongation, termination of polypeptide synthesis, translation in chloroplasts and mitochondria, post-translational modifications. Regulation of gene expression- regulation of gene expression in prokaryotes, concept of inducer and repressor, operons and transcriptional regulation (lac operon, tryptophan, arabinose operon, leucine responsive protein, sigma factors), post-transcriptional regulation, leader sequence and attenuators, alternative splicing, role of antisense RNA, gene regulation in bacteriophages, cro genes and lambda repressor protein, gene expression control in eukaryotes- role of specific DNA sequences, modification in DNA transcripts and histone proteins, regulation at translational level, phosphorylation of translation machinery, masking of mRNA, regulation by gene rearrangement, W-D repeat proteins. Genetic recombination, mutation and repair- genetic recombination between homologous DNA sequences, holliday model, role of Rec A protein, meiotic recombination (linkage and crossing over), bacterial transformation- transforming principle, competence, uptake and fate of DNA, conjugation- mechanism, role of plasmid, Hfr, transduction- lytic and lysogenic cycles, generalized and specialized transduction, transposition- transposable elements, insertion sequences, bacterial transposons, transposable elements in eukaryotes, molecular basis of mutation, types of mutations, spontaneous mutations, base analogues, tautomeric shift, frame shift mutations, chemical mutagens, intercalating substances, mutator genes, site-specific mutagenesis, mutational hotspots, methods of isolation of mutants, physical mutagens, biochemical mechanism of repair, photo-reactivation, excision repair, SOS repair.
9	**Membrane and ion channel biophysics:** Nature and magnitude of cell surface charge, electric properties of membranes, electric double layer, Poisson-Boltzmann theory of electric double layer, Guoy-Chapman model of electric double layer, influence of final size of molecule, and distribution of charge within individual molecules on the properties of electric double layer, influence of size and distribution of electrical charge of a membrane on transport of electrified molecules on the surface of a membrane, Hodgkin-Huxley equation, membrane impedance, relation between membrane potential and cell characteristics, Zeta, Stern and total electrochemical potential, Helmholtz-Smoluchowski equation, its correction by Debye-Huckle theory, thermodynamic and kinetic approaches to membrane potential, calculation of electrochemical potential by Nernst equation, transmembrane potential and its measurement by microelectrodes. Transport across biological membranes- osmosis and diffusion, electrostatic interaction between membrane surfaces, influence of components of solvents on the interaction between membranes, influence of electrical properties of molecules in solvents on the interaction between membranes, adhesion of membranes, diffusion, Flick's law, diffusion in two-compartment and multi-compartment systems, mechanisms of simple diffusion and facilitated diffusion, rate theory of membrane transport, electro-diffusion, osmosis, osmotic pressure, osmotic equilibrium, Donnan equilibrium, flow of water and solutes, electro-osmosis, molecular basis of aqueous channels, surface tension- definition, angle of contact, interfacial tension, capillary rise, determination of

	surface tension, temperature effect, preparation of liposomes and their utility, significance of surface tension in membrane biophysics. Active transport- selective permeability of membranes, selectivity and ion-specificity of biomembrane, ion channel structure and gating function, ion channel types and characterization, role of carriers in ion transport (valinomycin, gramicidin), Na-K ATPase, calcium ion transporting ATPase of sarcoplasmic reticulum, transport of macromolecules with and without vesiculation and by intermediate mechanism, transport and communication between cells and organelles, mechanisms of micro- and nano-vesiculation, influence of electrical properties of membranes and solvents on vesiculation of membranes, endocytosis, exocytosis, fusion of vesicles, encapsulation of nanoparticles and DNA, influence of detergents and nanoparticles on vesiculation and forms of membrane, mechanisms of stability and formations of membrane nanotubes and their role in transport of substances between cells and between cell organelles, microvesiculation of membranes and its role in spreading tumours and creation of blood clots, mechanisms of creation and stability of membrane pores.
10	**Biophysical techniques:** Hydrodynamic techniques- basic principles of centrifugation, forces involved, RCF, centrifugation techniques, their principles, types and applications, types of centrifuges, ultracentrifuge- design and types, differential centrifugation, density gradient centrifugation, preparative centrifuge, analytical ultracentrifuge, terminal velocity, sedimentation velocity, sedimentation equilibrium, viscometer, general features of fluid flow, nature of viscous drag for streamlined motion, viscosity coefficient of liquids, viscometric measurements, types of viscometers, relation between intrinsic viscosity and molecular weight, measurement of viscoelasticity. Physico-chemical fractionation- basic principles of adsorption and partition chromatography, types of adsorption and partition chromatography, applications of chromatography, paper chromatography, thinlayer chromatography, HPTLC, column chromatography, HPLC (gel filtration, ion exchange, hydrophobic interaction, affinity chromatography, normal phase, reverse phase, pumps, fraction collector), gas chromatography (gas-liquid, gas-solid chromatography), chromatographic columns for HPLC and gas chromatography, mobile phases for HPLC and gas chromatography, isocratic and gradient elution, efficiency of column, injection systems and detector for column chromatography, ion chromatography, membrane techniques for separation-equilibrium dialysis, ultra-filtration and other membrane techniques. Electroanalytical techniques- electrophoresis principle, electrophoretic mobility, factors affecting electrophoretic mobility, electrophoresis set-up, design, vertical and submerged electrophoresis, free electrophoresis, paper electrophoresis, starch gel electrophoresis, agarose gel electrophoresis, native PAGE, SDS-PAGE, pulsed-field electrophoresis, capillary electrophoresis, isoelectric focusing, 2D electrophoresis, gradient gel electrophoresis.
11	**Nanobiophysics** Origins of nanotechnology, nanoscience, the unit nanometer. Nanobiomaterials (lipoprotein based, liposomes, peptide-based). Self-assembled nanomaterials. Physico-chemical properties of nanomaterials, optical, mechanical, electrical, magnetic, thermal properties etc. Nanomotors in biological system: ATP synthase, flagellar motor in bacteria, kinesin, myosins, dynein etc. Concept of bottom- up approach, Top-down approach. Principles of gas, liquid & solid phase synthesis of nanomaterials. Biological methods of synthesis of nanomaterials. Synthesis of various nanomaterials from bacteria, fungi, plants, algae, DNA & protein templates etc. Physical and chemical methods to synthesize nanomaterials.Characterization techniques for nanomaterials using spectroscopic and microscopic techniques (UV-visible, fluorescence, light scattering, Force microscopy, Transmission and Scanning electron microscopy, High resolution Transmission Electron Microscopy, X-ray diffraction, Ellipsometry, surface tensiometry etc) Physicochemical properties of Nanoparticles influencing their toxic effects on biological system, cell, tissue and organs, mechanism related to nanotoxicity. Interaction of nanomaterials with biological system i.e. cell, cell membranes, tissues, organ, protein, DNA, RNA and enzymes. Pharmaceutical development based on cytotoxicity of nanomaterials, in vitro testing methods for nanomaterial toxicity, Nanomaterials and immune system, biocompatibility and stability. Health effects of nanomaterials.

12	**Aviation, high altitude, space and deep sea physiology** Aviation, high altitude physiology- effect of low oxygen pressure on human body, mountain sickness, high altitude physiology including respiratory physiology in flight, hypoxia and its prevention, oxygen systems, decompression sickness, rapid decompression, aero-medical problems in high altitude operations, hyperbaric oxygen treatment, environmental physiology including thermal stress, thermal regulation and prevention of thermal effects. Space physiology- effects of acceleratory forces on the body in aviation and space physiology, radiation and temperature, problems at high altitude and space, weightlessness in space, physiological adaptation to space flight, cardiovascular and respiratory changes during acceleration, protection against +Gz, effects of –Gz, transverse and lateral G, fundamentals of space physiology simulation of micro G, conditioning, training and evaluation, life support systems, escape and survival, space sickness, countering gravitational and thermal stresses. Deep sea physiology- concept of deep sea environment, diving at ambient pressure, physiological effects of high external pressure different external systems of human body, decompression sickness- pathophysiology and prevention, preventive measures for hypo- and hyperbaric pressure, physiology in deep sea diving and other high pressure operations, Caissons disease, cyanosis, dyspnoea, hyperopnea, apnoea and asphyxia.
13	**Bioinformatics & Structural Biology** Information theory and Bioinformatics Network- Information theory, Relation between information & entropy, Redundancy theorem & noise, Information content of biological system, Biological data exploration through internet Resources EMB net, NCBI, BTIS network, Bioinformatics landscape intrinsic & extrinsic view, Cheminformatics & medical informatics. Biological databases: Sequence databases, Protein sequence databases, Structural databases, PDBs, Motif databases, Protein motif database, Genome databases, Proteome databases etc. Genomics and Proteomics- Genome information resources, Functional Genomics DNA sequence analysis, Gene bank, CDNA library, pharmaco-genomics, ESTs analysis method for recognition of functional signals, Consensus sequences, approaches to gene identification using internet resources, Concept & applications of DNA microarray technology, Protein sequences information & features, Proteomic analysis using internet resources, Prediction of protein structure, Protein folding, Problem & functional sites, Phylogeny, Methods of phylogenetic analysis, Application of sequence analysis & phylogenetic information in diversity studies. Introduction to Next-Generation sequencing (NGS); Overview, history and development of sequencing techniques; Basic terms and principles; NGS applications and workflow; NGS technologies and platforms, Library preparation; Standard applications and techniques. Bioinformatics tools- Pair wise Alignment, Alignment algorithms, sequence analysis tools, BLAST (Basic Logical Alignment Search Tool) FASTA, Multiple Alignment, Sequence analysis using EMBOSS. Molecular Modelling: Introduction to computer graphics, Visualization of bimolecular structures, concepts in molecular modelling. Energy minimization, Dynamic stimulation & conformational analysis, Applications of molecular modelling packages, structural similarity & overlaps, structural prediction & molecular docking. Applications of protein modelling.
14	**Genetic Engineering** Genetic Recombination in Prokaryotes & Eukaryotes- Transformation- transforming principle, competence, uptake and fate of DNA, Conjugation- mechanism of conjugation, Role of F plasmid, Hfr. Transduction-lytic and lysogenic cycles, generalized and specialized transduction. Transposition-Transposable Elements, Insertion Sequences, bacterial transposons. Concept of Recombinant DNA and Steps in gene cloning, Enzymes-Restriction endonucleases and their restriction sites, ligases, polymerases, other DNA modifying enzymes, Vectors-properties of good vector, Plasmid vectors- pBR322, pUC18/19, pGEM3z, A phage vectors, Phagemid vectors, Artificial Chromosome Vectors-BAC, YAC, Shuttle vectors, Vectors for animals & plants. Techniques in Recombinant DNA Technology- Enzymatic synthesis of DNA-PCR: Denaturation, PCR primers, annealing, primer extension, DNA sequencing, Nucleic Acid blotting, Southern blotting, Northen blotting, Western blotting, Colony Hybridization and Dot-blot technique.

	Gene transfer methods in animals and plants- Agrobacterium mediated transformation (Ti & Ri plasmid-based vectors of Agrobacterium), Transfection with modified viral vectors Physical delivery methods microinjection, macro injection, microprojectile & electroporation; Chemical methods -DNA use of polyethylene glycol, Calcium phosphate, DEAE Dextrose, Use of polycation DMSO etc. Applications of Genetic Engineering- Agriculture- Transgenic Plants for disease resistance, protein production, herbicide resistance, Delayed ripening, Transgenic animals- as bioreactors for the production of regulatory proteins, vaccines and hormones. Transgenic and gene knockout technologies to study molecular biology and oncology, Stem Cell Therapy, Gene therapy in Cystic Fibrosis DMD, SCID. RNAi as a tool of gene therapy, Industrial Applications- Production of recombinant therapeutic proteins- r-insulin, r-hGH, r- Erythropoietin, r-Factor VIII & IX, r-Hepatitis B vaccine, Construction of Genomic Library and cDNA library.
15	**Photo-biophysics** Biophysical Ecology- Introduction to Ecosystem Micro climate & energy environment, Influence of physical factors, Interaction between environment & Biosystems, solar radiation, Photochemical filtering of solar radiation. Atmospheric absorption, spectrum & thermal emission spectra, atomic scattering, Comparative distribution of natural light, spectral properties of liquid water, plant & animals, Green house effect, Non-ionizing & Ionizing radiation. Photophysics& Photochemistry- Nature and measurement of light, Physical properties of excited molecules, Phophysical processes, fluorescence, Photophosphorescence, Internal conversion, Intersystem crossing, Photophysical spectra, Actionspectra, Optical activity, Photophysical kinetics of bimolecular processes. Basic principles and laws of photochemistry, Quantum photochemical principles, Photochemical primary processes, Types of photochemical reaction, Photochemistry of amino acids and proteins, Photochemistry of DNA & RNA and its constitutes, Recovery from photochemical damage, Photophysical and photochemical aspects of photosensitization, Chemiluminisence, Mechanism and significance, Techniques for study of transient species in photochemical reaction. Photobiological phenomenon & Circadian Rhythm- Photoactivation of biological systems, Photodynamic dyes and mechanism of photodynamic action on cells, Viruses, Proteins and nucleic acids, Concepts, Mechanism and Significance of photomorphogenesis, Photoperiodism, Phototaxis, Phototropism, Photosynthesis, Light acceptor, system, Photosystem as Photosynthetic reaction centre, Photophosphorelation, Bioluminescence, General failures of circulation rhythms, Entrainment to environmental cycles, Mechanisms of circadian rhythms, Circadian organization in multicellular organism including human.
16	**Biophysical & Bioanalytical Techniques.** Fluorescence & Optical spectroscopy- Fluorescence spectroscopy: Principle, Instrument Design FRET, & Applications- Fluorescent probes, modification in methionine, histidine, tryptophan, amine and carboxylic groups, fluorescence life-time and quenching studies & applications in proteins & membrane studies. Energy transfer for distance measurement in proteins & membranes. Use of fluorescence polarization and anisotropy, measurement of anisotropy decay. Comparative study of rigid proteins i.e.; lysozyme and lactalbumin. Internal flexibility of multidomain proteins, i.e. myosin, fraction, fibrinogen. Fluorescence dye-Nucleic acid complexes. Principle and Instrument Design of Optical spectroscopy, Methods & Applications of Polarimetry, Light scattering, Refractometry, Circular dichroism (CD), optical rotatory disperson (ORD): Plain, circular and elliptical polarization of light, Absorption by oriented molecules, Relation between CD and ORD, Dichroic ratio of proteins and nucleic acids. application of ORD in conformation and interactions of biomolecules, Determination of structural correlations in biomolecules using CD & ORD,. Relationship between molar ellipticity of CD, Conformational dependence of CD helical structure, coupling between chromophore etc. Secondary and tertiary structures of peptides and proteins, effect of pH, temperature, organic solvents and neutral salts. Conformational information- aromatic and disulphide side chains. CD spectra of di, oligo and polypeptides, structure of supra- molecular structure i.e. membranes and ordered aggregates of chromophore. ESR Spectroscopy- Magnetic moment of unpaired electrons and paramagnetic resonance,

	Principle of operation and working of electron spin resonance, E.S.R. spectrometer, Hyperfine ESR spectroscopy, representation of ESR spectrum, E.S.R. spectra of organic radicals in solution isotropic hyperfine splitting, ESR spectra of organic radicals in solids-anisotropic hyperfine splitting. ESR spectra of inorganic radicals-g-value anisotropy. ESR of organic molecules in triplet states-Electron spin-spin interactions. Relaxation processes and line shapes, 'g' -value, spectra of simple organic free radicals, hyperfine coupling, prediction of expected number of lines and intensities. Spectra of transition metal complexes, Zero-field splitting, utility for identification of radical; spin labelled probes, spin-labelling: A reporter group technique, requirement of such a group, Nitroxides spin label probes and their molecular structures, Anisotropy of the value order parameters, information obtained from ESR motion, polarity, biochemical data, orientation Intra- molecular distances etc. Applications of these concepts to (i) studying the structure and function of enzyme.. i.e. lysozyme etc. (ii) conformational change of molecular artifact in try spin, spin labelled ligands as probe for rigidity of binding sites, lipid spin labels in the biological membranes etc. applications in biology, pharmacy. Mass Spectroscopy- Basic principles, brief outline of instrumentation, fragmentation processes & ion formation, molecular ions, meta-stable ions, fragmentation patterns and fragment characteristics in relation to parent structure and functional groups, Detectors & their designs, relative abundances of isotopes and their contribution to characteristic peaks, mass spectrum; its characteristics, presentation and interpretation, chemical ionization mass spectrometry, GC- MS including recent advances in MS, Fast atom bombardment mass spectroscopy; analysis of drugs in biological samples by combined GC-MS. Chemical ionization mass spectroscopy (CIMS), Field, Ionization Mass Spectrometry (FIMS), Fast Atom Bombardment MS (FAB MS), Matrix Assisted laser desorption / ionization MS (MALDI-MS), interpretation of spectra and applications in biology, Pharmacy etc
17	**Radiation Biophysics** Radiological Physics- Atomic structure and atomic nuclei, Radioactivity, Isotope, laws of Radioactivity, Alfa, Beta, Gamma rays, Properties of Electromagnetic radiation, Radiation Units-Units of radioactivity, exposure & dose, Dose equivalent Unit, X & Gamma ray interaction with matter (Photoelectric & Compton effect, Ion pair production Heavy charged particles & Neutrons, linear, mass, electronic & atomic attenuation coefficient-, HVL, Mean free path, Absorption edges, LET. Radiochemistry and free Radicals- Radiolysis of water, Production of free radicals & their interactions, Pulse radiolysis, Role of scavengers, G-value, Direct and Indirect action, Oxygen and temperature effect, OER, Target theory, Single & Multi hit theory, Multi target theory, Calculation of target, Mass, Volume & Molecular weight, Effect of radiation on Nucleic acids, Proteins, Enzymes & Carbohydrates, Cellular effects of radiation, Mitotic delay, Inhibition of mitosis, Giant cell formation, Cell death, Cell recovery & Modification of Radiation damage, Genetic Effect of radiolysis, Factors affecting frequency of radiation induced mutation, Chromosomal breakage and Aberrations, Somatic effect of radiation, Physical factors influencing somatic effects, Dependence on dose, Dose rate, Acute radiation damage, LD-50, Radiation syndrome, Early and late effects of radiation, Effect of Chronic exposure to radiation, Dose effect relationship, Genetic burden, Concept of doubling dose & its effect on genetic equilibrium. Radiation detection, Measurement and Safety measures- Principles of radiation detection and measurement, General requirements of Dosimeters, Radiation sources, Tele gamma Unit (Cobalt Unit), Gamma chamber, Nuclear reactors, Thermal & fast neutron sources, Basic principles, Design & Working of physical dosimeters- Ionization chamber, Proportional counters, GM-Counter, Concepts of Gas amplification, Resolving time & Dead time, Scintillation Detectors, Thermolumeniscent Dosimeter, Semiconductor, Surface barrier & Lithium detectors, Area survey meter & Pocket dosimeter, Film badge, General principle of chemical dosimetry, Salient Features of Chemical dosimeter, Dose evaluation formula for chemical dosimetry, Principles of radiolytic reaction, Experimental methods- Influencing factors of Fricke dosimeter methyl orange, FBX dosimeter, Free radical dosimeter, Ceric sulphate dosimeter, PMMA, PVC, chlorobenzene dosimeter, High & low dose indicators. Natural & Man-made radiation exposures or Principles of dose equivalent limit (DEL) radiation protection, Maximum permissible dose (MPD), Evaluation of external & internal radiation hazards, Radiation protection measures in industrial establishment, Radioisotope labs, diagnostic

	& therapeutic installation & during transportation of radioactive substances, disposal of radioactive waste, administrative & legislative aspect of radiation protection.
18	**Medical Biophysics** Electrophysiology- Principles of Electrocardiography, Heart- an electric potential sources, ECG waveforms, Standard lead systems, ECG preamplifiers, ECG readout devices, ECG machine, Measurements, Trouble-shooting, Principles of Electroencephalography, EEG Electrodes, 10- 20 Electrode system, EEG Amplitude & Frequency band, Multichannel EEG recording. EEG in Sleep, Diagnostic Application of EEG, Recording of visual & auditory evoked Potentials, EEG Telemetry system, EEG System artifacts, Faults, Troubleshooting & Maintenance, Other electrophysiological recordings, EMG, ERG, FOG & their applications. Medical-Imaging Techniques & Nuclear Medicine- Physical aspects of Medical-imaging, Principle, Practical System, Medical utility of X-ray imaging, Mammography, Xeroradiography, Fluoroscopy, Computerized Axial Tomography, Angiography, Myelography, Magnetic resonance imaging, Ultrasonography. Basic principles of Nuclear Medicine, Diagnostic use of Radioisotopes In-vivo & In-vitro procedures, (Single isotope, Double isotope methods), Radio immunoassay counting system, General principles & procedures of organ scanning. Renal imaging, Cardiac imaging. Thyroid scanning. Blood volume determination by isotope method, Rectilinear scanners & Gamma scintillation camera, Positron emission Tomography (PET), Single Photon emission computer Tomography (SPECT), Radio pharmaceuticals & their Diagnostic applications. Radiotherapy- Concepts of teletherapy & Brachytherapy, Co-60 Therapy, Basic principles & scope of radio therapy, Benign & Malignant tumors, Tissue tolerance dose &Tumor lethal dose, Medical dosimetry. Dose fractionation, Palliative & Curative therapy. Treatment planning, Isodose distribution, Patient data, Correction & Setup, Field shaping, Skin dose and field separation, brachytherapy, Sources, Calibrations, Dose distribution implant dosimetry.
19	**Genomics & Proteomics** Introduction to Genomics and Proteomics- Overview research areas and related journals in genomics and proteomics - Concepts of central dogma Structure and organization of prokaryotic & eukaryotic genome, organization and structure of genome, genome size, sequence complexity, introns and exons. Genome structure in viruses, bacteria, eukaryotes. Changes and regulation of genome activity in prokaryote and eukaryote, Genetic maps-Linkage maps, Cytogenetics maps & Physical maps. Brief outlook of various genome projects and their outcomes Human genome project, Genome mapping and sequencing mapping techniques Genetic markers RFLP, SSLP, STRS, VNTRS, Sequencing methods: chemical and enzymatic method, high throughput method. Automated sequencing methods - Whole genome shotgun sequencing method, NGS sequencing method. Gene identification and Expression profiling- Gene annotation, traditional route of gene identification, detecting ORF, gene ontology. Overview of comparative genomics, global expression profiling- traditional approaches to expression profiling, analysis of RNA expression, application of genome analysis, application of functional genomics, gene knockdown, gene editing Cripsr- Cas system, Sequence assembly and annotation, Assembly of contiguous DNA sequence- shotgun, directed shotgun and clone contig approach, Tools for sequence assembly, structural and functional genomics- Transcriptomes, Pharmacogenomics. Genome variation: variations in human genome, known examples of variants that cause disease, pharmacogenomics, ethical consequences of genomic variations. Expression data analysis: DNA/RNA microarrays, oligomicroarray, chip technology, Affymetrics protocol, data generation, spotted microarray technique, biomedical applications. Proteome analysis- Introduction to proteome: Proteome and technology, Importance of 2D Electrophoresis in proteomics - Protein identification in proteome projects - Primary and secondary attributes for protein identification, Detection and analysis of co- and post-translational modification, protein identification (2 D electrophoresis), mass spectroscopy based methods for protein identification, MALDI TOF, application of NMR, mining protein databases, application to human disease studies, Proteome databases: Protein sequence databases SWISS-PROT and TREMBL Pattern and profile databases PROSITE and BLOCKS 2D PAGE databases Structure databases - PDB- Metabolic databases post translational modification databases Application of proteomics to medicine, proteomics, toxicology and pharmaceuticals.

20	**Cellular and Molecular Neurophysiology** Neurophysiology and Biophysics- Basic concepts in neurobiology and biophysics, Historical perspectives and key discoveries, Neurons and Neural Networks Structure and function of neurons, Ion channels and membrane potentials, Synaptic transmission and neural circuits, Biophysical Techniques in Neuroscience, Electrophysiology (patch-clamp recording, voltage-clamp techniques), Imaging techniques (fluorescence microscopy, calcium imaging), Computational modelling of neural processes. Molecular Neuro-biophysics- Neurotransmitters and Receptors, Neurotransmitter types (e.g., glutamate, GABA, acetylcholine), Receptor subtypes (ionotropic, metabotropic), Ligand-gated ion channels. Neuronal Excitability and Action Potentials, Hodgkin-Huxley model, Voltage-gated ion channels (sodium, potassium), Propagation of action potentials, Synaptic Plasticity and Learning Long-term potentiation (LTP) and long-term depression (LTD) Role of calcium signalling, Molecular mechanisms underlying synaptic plasticity. Neurodegenerative disorders- Neural Oscillations and Synchronization, Brain rhythms (alpha, beta, gamma), Synchronization in neural networks, Role of oscillations in cognition and behaviour, Neuro- biophysics of Sensory Systems, Vision (photoreceptors, visual processing), Audition (hair cells, auditory pathways), Somatosensory systems (mechanoreceptors, touch), Neurological disorders and Therapeutics, Neurodegenerative diseases (Alzheimer's, Parkinson's), Drug targets and therapeutic interventions, Challenges in drug delivery to the brain.

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग